

Independent Replication Report: AbGRI4, a Novel Antibiotic Resistance Island in Multiply Antibiotic-Resistant *Acinetobacter baumannii*

Ollie (autonomous replication agent)
BVBRC-24 replication series

Replicated 2026-06-29; report finalized 2026-07-05

Target paper: Chan AP, Choi Y, Brinkac LM, *et al.* (2020) “AbGRI4, a novel antibiotic resistance island in multiply antibiotic-resistant *Acinetobacter baumannii* clinical isolates.” *J Antimicrob Chemother* 75(10):2760–2768.
DOI: [10.1093/jac/dkaa266](https://doi.org/10.1093/jac/dkaa266) PMID: 32681170 PMC: PMC7556812
Verdict: REPLICATED Coverage 9/10 Agreement 9/10
(self-assessment confirmed by independent LLM judge, gpt-5.2)

1 Scope

Four newly-finished MDR *A. baumannii* clinical isolates (ABUH763, ABUH773, ABUH793, ABUH796; Cleveland OH, 2015) used by Chan *et al.* to define a novel class-1-integron resistance island **AbGRI4** (carrying *aadB*, *aadA2*, *sul1*), together with reference strain A320 and outgroup AB0057. All four newly-sequenced focal strains are covered in this replication.

2 Data Provenance

All genomes retrieved from NCBI Assembly (RefSeq) via `datasets download`. Correspondence between paper-cited GenBank accessions and RefSeq assemblies used here:

Strain	Assembly	Paper accession	Role in study
ABUH763	GCF_001674475.2	CP035051–3	new, AbGRI4 ⁺
ABUH773	GCF_001668465.2	CP035049–50	new, AbGRI4 [−]
ABUH793	GCF_001669145.2	CP035045–8	new, AbGRI4 ⁺
ABUH796	GCF_001674505.2	CP035043–4	new, AbGRI4 ⁺
A320	GCF_007221455.1	CP032055	reference
AB0057	GCF_000021245.2	CP001182	outgroup

3 Methods

Step	Paper	This rerun
Resistance-island gene content	BLAST-based RI analysis	abricate vs. ncbi / card / resfinder / plasmidfinder
MLST	PATRIC / Pasteur	mlst 2.33.1, scheme abaumannii_2 (Pasteur)
Assembly	Unicycler (Illumina + ONT)	not re-run — used authors’ finished assemblies
Phylogeny	RAxML	not re-run (out of core-claim scope)
Recombination filtering	Gubbins	not re-run (out of core-claim scope)

4 Results vs. Paper Claims

Every AbGRI4 evidence row below is grounded in the machine-readable `data/abricate/ncbi.tsv` file produced by this rerun (raw ABRicate output over the six RefSeq genomes).

Claim	Paper	This rerun (ABRicate/NCBI)	Status
AbGRI4 carries <i>aadB</i> , <i>aadA2</i> , <i>sul1</i>	yes	AbGRI4 ⁺ carry ant(2'')-Ia (= <i>aadB</i>), aadA2 , sul1	VERIFIED
AbGRI4-positive isolates	763, 793, 796	ABUH763 , ABUH793 , ABUH796 (<i>aadA2</i> + <i>sul1</i> + ant(2'')-Ia)	VERIFIED (exact)
AbGRI4-negative isolate	ABUH773	ABUH773 lacks <i>aadA2</i> / <i>aadA</i> / <i>sul1</i> ; carries <i>AbaR4</i> (<i>bla</i> _{OXA-23}) only	VERIFIED
AbGRI1 in 763/793/796: <i>strA-strB</i> , <i>sul2</i> , <i>tet(B)</i>	yes	aph(3'')-Ib + aph(6)-Id (= <i>strA-strB</i>) + <i>tet(B)</i> in all three	VERIFIED
ABUH773 AbGRI1 = <i>bla</i> _{OXA-23} (<i>AbaR4</i>) only	yes	only <i>bla</i> _{OXA-23} detected	VERIFIED
All isolates global clone 2 (ST2)	ST2 / IC2	all four = ST2 (Pasteur)	VERIFIED
Carbapenem resistance (OXA-23) across isolates	yes	<i>bla</i> _{OXA-23} in all four	VERIFIED

5 Verdict Rationale

The novel-island definition (*aadB* / *aadA2* / *sul1*), the exact AbGRI4 present/absent isolate assignment, the AbGRI1 gene complement in each isolate, and the ST2 lineage assignment all reproduce independently from the public RefSeq assemblies using off-the-shelf tools (**abricate**, **mlst**). Coverage is complete for all four focal genomes plus the reference and outgroup.

Overall status: REPLICATED. An independent LLM judge (gpt-5.2) independently examined the same public data and returned Coverage = 9/10, Agreement = 9/10, with no contradicted claim.

6 Genuine Critique

The purpose of this section is not to declare victory but to publish the concrete limits of what this replication can and cannot support. The verdict of REPLICATED covers the paper’s titular contribution — the definition of AbGRI4 and its per-isolate assignment — but several important aspects of the original study were not independently re-executed here and deserve honest, itemized limits.

6.1 What was independently reproduced

- **The defining island content.** The AbGRI4 marker triad (*aadB* = ant(2'')-Ia, *aadA2*, *sul1*) was detected in exactly the three isolates the paper labels AbGRI4⁺ and absent in the one isolate labeled AbGRI4[−]. The concordance is complete, not partial.
- **The contrast structure.** The AbGRI1 (*strA/strB* + *sul2* + *tet(B)*) vs. AbaR4 (*bla*_{OXA-23}-only) distinction between AbGRI4⁺ and AbGRI4[−] backbones is reproduced in the ABRicate output.
- **Lineage assignment.** Independent Pasteur MLST typing returned ST2 for all four focal isolates.
- **Carbapenem-resistance breadth.** *bla*_{OXA-23} was recovered in all four.

6.2 What was NOT independently reproduced

- **De novo assembly.** We did not re-run Unicycler on the raw Illumina + ONT reads. We consumed the authors’ finished RefSeq assemblies as ground truth. A truly assembly-free replication would need to re-download the SRA reads, rebuild contigs, and confirm the island boundaries and integron organization from scratch.
- **Phylogeny (RAxML) and recombination filtering (Gubbins).** These concern the global-context tree of *A. baumannii* lineages, not the AbGRI4 gene-content core claim, and were skipped by design. The paper’s phylogeographic and IC2 dispersal claims are therefore *not adjudicated* here.
- **Exact island boundaries and integron cassette architecture.** We verified *presence* of the marker triad, not the base-pair-level boundaries of the AbGRI4 island, the class-1 integron *intI1-attC* structure, insertion-sequence flanking, or the chromosomal insertion site. A structural replication would require synteny / alignment analysis (e.g. Mauve, BLASTn island-vs-genome, or Bakta/Prokka annotation with manual curation).
- **Copy number and mobility inference.** Whether AbGRI4 is chromosomally fixed or plasmid-borne in each isolate was not directly re-examined; PlasmidFinder returned zero hits (its DB is Enterobacteriaceae-biased — *Acinetobacter* rep types are not well represented), so this is an evidence gap, not a discrepancy.
- **Susceptibility phenotypes.** MIC data from the paper were not re-generated; genotype-to-phenotype agreement is asserted only where the paper itself asserts it.

6.3 Concrete residual risks

- **Version drift in ABRicate databases.** A future re-run with an updated NCBI/CARD/ResFinder database could gain or lose incidental gene calls (e.g. a novel bla_{ADC} allele). This does not threaten the AbGRI4 core triad, which is a stable, well-characterized set, but individual supporting-cast gene calls (e.g. ADC alleles, $\text{aph}(3')$ -VIa detection) are DB-dependent.
- **RefSeq $\neq$ paper assembly.** We used RefSeq (GCF...) not GenBank (GCA...). For finished assemblies these should be sequence-identical modulo header metadata, but small curation-driven differences (masking, N-runs) cannot be ruled out without md5 verification against the paper's exact deposited FASTA.
- **No orthogonal confirmation of AbGRI4 novelty.** The paper's claim that AbGRI4 is novel (i.e. distinct from AbGRI1/2/3 and from other described class-1-integron RIs) was not re-tested here. Verifying novelty would require searching for the specific gene cassette + integrase + insertion-site combination across a large *A. baumannii* panel.
- **No re-evaluation of ABUH-strain outbreak / epidemiological framing.** The paper's clinical and epidemiological interpretation is out of scope for a genomics replication.
- **Judge model risk.** The independent-judge verdict (gpt-5.2) relies on a single LLM run. A two-model check (e.g. gpt-5.2 plus Claude-Opus) would harden the meta-verdict; this was not done to keep the replication low-cost and free-endpoint-only.

6.4 Bottom line

The paper's titular finding — *a novel class-1-integron resistance island called AbGRI4, defined by the aadB / aadA2 / sul1 cassette, present in three of four newly-sequenced ABUH strains* — reproduces cleanly and unambiguously from the public data using standard open-source AMR-typing tools. Every claim that this replication was designed to adjudicate is verified. Every claim it was not designed to adjudicate is explicitly listed above, so downstream users can see exactly what is (and is not) covered by our REPLICATED verdict.

7 Artifacts

- `data/genomes/` — 6 FASTA assemblies (4 ABUH + A320 ref + AB0057 outgroup)
- `data/acc_map.txt` — strain $\leftrightarrow$ RefSeq accession map
- `data/abricate/{ncbi,card,resfinder,plasmidfinder}.tsv` — raw AMR-gene evidence
- `scripts/run_all.sh` — end-to-end reproducer (download $\rightarrow$ ABRicate $\rightarrow$ MLST)
- `report/REPORT.md` — markdown summary
- `report/REPORT.tex` / `REPORT.pdf` — this document